1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

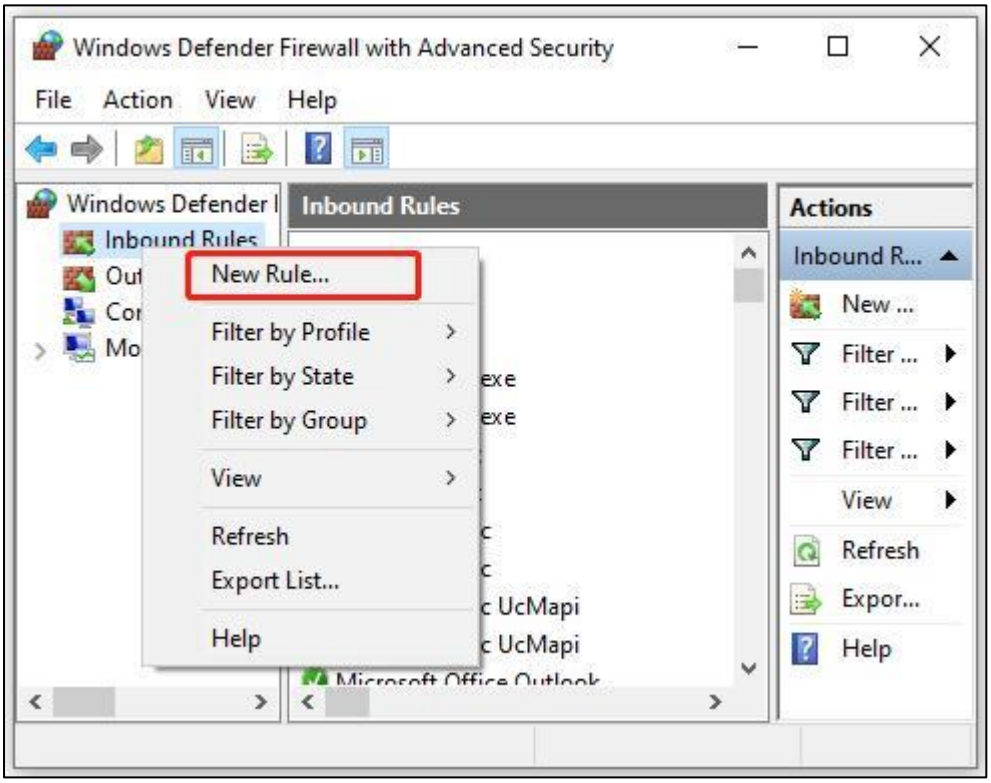
Step 1: Open Windows Defender Firewall

1. Press **Windows + R** on your keyboard.
2. Type wf.msc.
3. Press **Enter**.

The **Windows Defender Firewall with Advanced Security** window will open.

Step 2: Create a New Inbound Rule

1. Click **Inbound Rules** in the left-side panel.
2. In the right-side panel, click **New Rule...**.

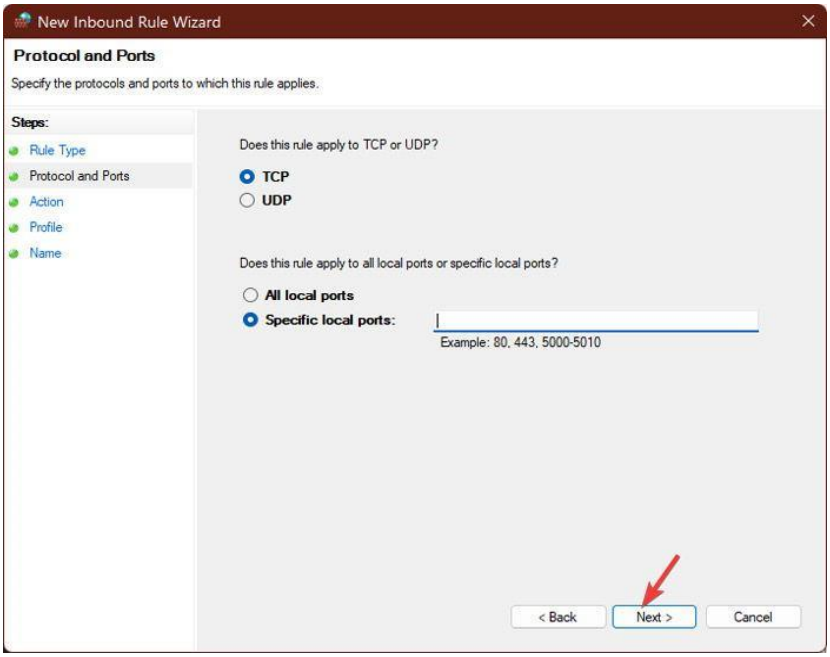


1. In the Rule Type window, select **Port**.
2. Click **Next**.

Step 3: Configure Port 8080

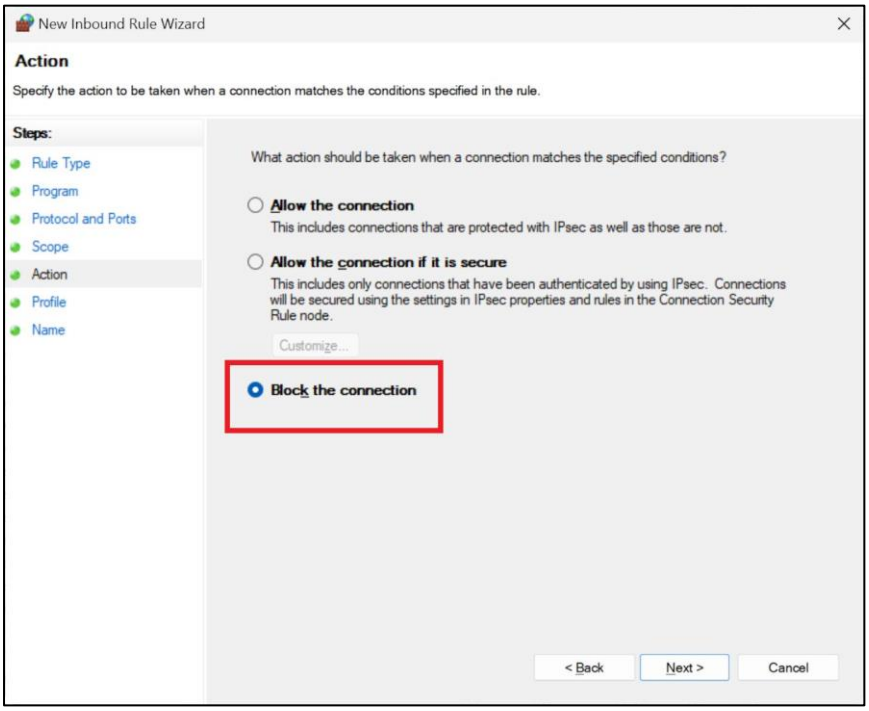
On the Protocol and Ports screen:

1. Select **TCP**.
2. Select **Specific local ports**.
3. Enter 8080 in the box.
4. Click **Next**.



Step 4: Block the Connection

1. Select **Block the connection**.
2. Click **Next**.



Step 5: Apply the Rule to All Profiles

Select all three checkboxes:

- Domain
- Private
- Public

Click **Next**.

Step 6: Name the Rule

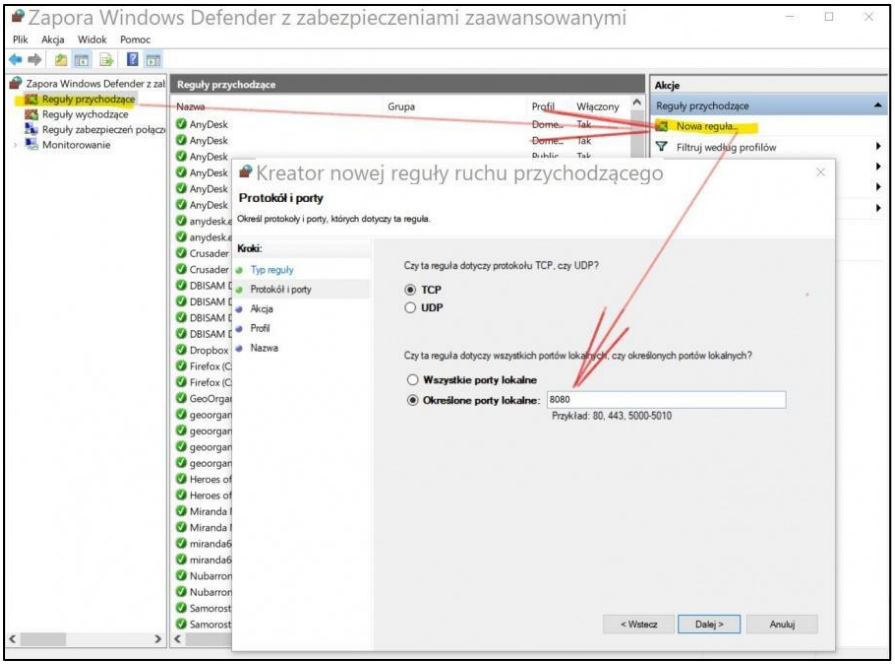
Enter the following details:

Field	Value
Rule name	Block Incoming Port 8080
Description	Blocks incoming TCP traffic on local port 8080

Click **Finish**.

Your new inbound rule should now appear in the Inbound Rules list.

Step 7: Take a Screenshot



- Open **Windows Defender Firewall with Advanced Security**.
- Click **Inbound Rules**.
- Find Block Incoming Port 8080.
- Make sure the rule is enabled.
- Press **Windows + Shift + S**.
- Select the area containing your rule and save the screenshot.

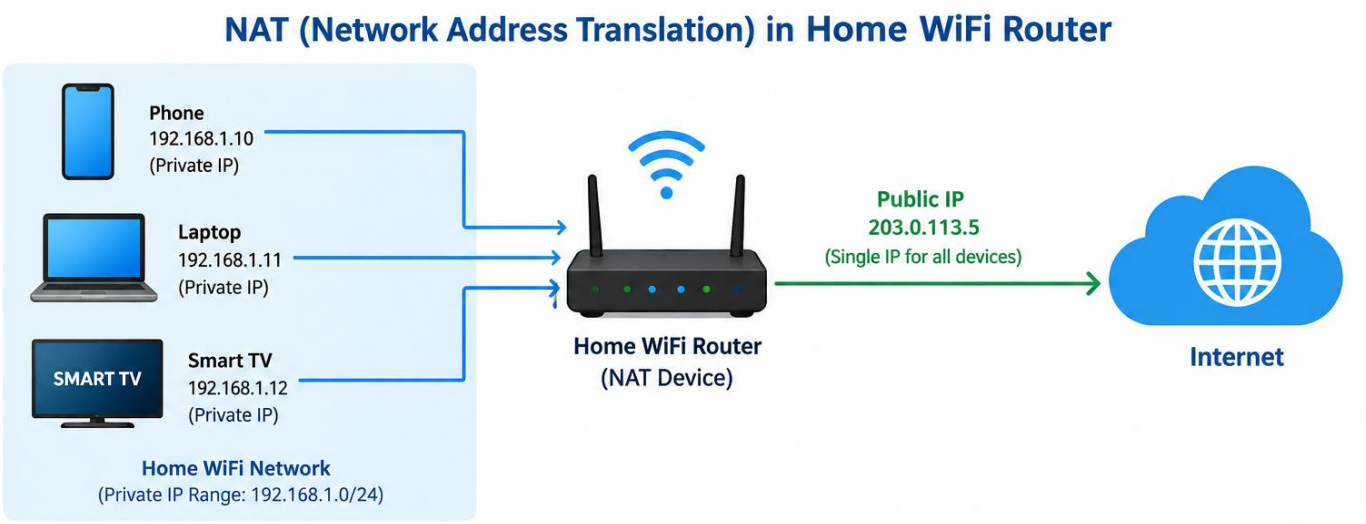
Step 8: Verify the Rule

Double-click your new rule and check that:

- **Action:** Block the connection
- **Protocol:** TCP
- **Local port:** 8080
- **Enabled:** Yes

2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.



How it works?

(a) Your device (Phone, laptop or tv) use private IP addresses (192.168.1.x) inside your home network.
(b) The Router performs NAT (Network Address Translation) and changes the source IP to your public IP when sending data to the Internet.
(c) Return traffic from the Internet is mapped back to the correct device using a NAT table.

3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Step 1: Open Your Router's Admin Panel

1. Connect your laptop to your home Wi-Fi.
2. Open Chrome and enter your router's gateway address, commonly 192.168.1.1 or 192.168.0.1.
3. Log in using your router's administrator credentials.

Step 2: Find Port Filtering Settings

Look for a menu named Security, Firewall, Access Control, or Port Filtering. The exact name depends on your router model.

Step 3: Add the Port Filtering Rule

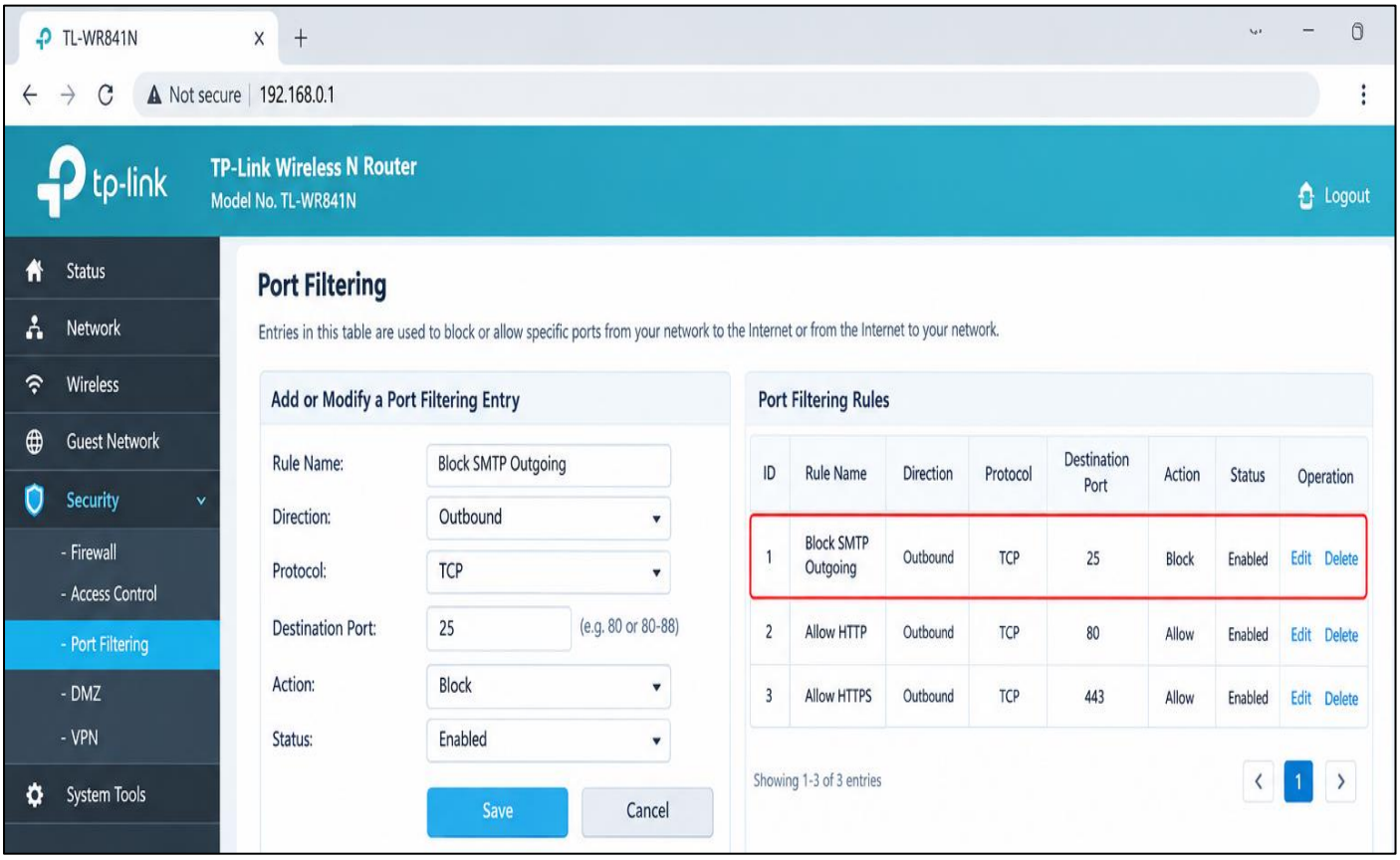
If your router supports outbound port filtering, enter these settings:

Setting	Value
Rule name	Block SMTP Outgoing
Direction	Outbound
Protocol	TCP
Destination port	25
Action	Block / Deny
Status	Enabled

Click Apply or Save to activate the rule.

Step 4: Explain How Blocking Port 25 Prevents Spam

Blocking outgoing TCP port 25 prevents devices on your network from directly connecting to external SMTP servers on that port to send email. This can reduce spam sent by malware-infected devices, although it does not prevent all spam because attackers may use alternative ports or email services.



4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.

To determine which packets are allowed or blocked, we need to compare each packet's source IP, destination IP, and destination port against the firewall's packet filtering rules.

Here is an example using a sample rule set.

Sample Packet Filtering Rule Set

Rule	Protocol	Destination Port	Action
1	TCP (HTTP)	80	Allow
2	TCP (SSH)	22	Block
3	UDP (DNS)	53	Allow
4	All other traffic	Any	Block

Packet Analysis

Packet	Source IP	Destination IP	Destination Port	Decision
1. HTTP request	192.168.1.5	172.217.166.78	80	Allowed
2. SSH request	10.0.0.10	34.201.23.1	22	Blocked
3. DNS query	192.168.1.7	8.8.8.8	53	Allowed

5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Instagram and Zomato are popular online platforms that handle millions of requests. Firewalls and packet filtering can help protect their servers from cyberattacks and unauthorized access.

1. Block Unauthorized Access

Instagram and Zomato can use firewalls to allow legitimate network traffic and block unauthorized connections. For example, they can allow HTTPS traffic on port 443 while restricting access to internal database ports to protect sensitive information.

2. Protect Against DDoS Attacks

Attackers may send a large number of requests to overload a server. Firewalls and DDoS protection systems can identify and filter malicious traffic, helping Instagram and Zomato keep their services available to legitimate users.

3. Block Malicious Requests and Bots

Both platforms can use Web Application Firewalls (WAFs) to detect suspicious requests, malicious bots, and attacks such as SQL injection. They can also apply rate limits to reduce automated abuse of login pages, APIs, and other services.

Conclusion

Firewalls and packet filtering help protect servers by blocking unauthorized traffic, reducing the impact of DDoS attacks, and filtering malicious requests. These measures improve service availability and help protect user data.